

Imperial College London

MSc EXAMINATION June 2024

STATISTICAL METHODS FOR EXPERIMENTAL PHYSICS

For MLBD MRes Students

June 5th, 2024: 10:00 to 12:00

Answer all FOUR questions

Marks shown on this paper are indicative of those the Examiners anticipate assigning.

General Instructions

Complete the front cover of each of the 4 answer books provided.

If an electronic calculator is used, write its serial number at the top of the front cover of each answer book.

USE ONE ANSWER BOOK FOR EACH QUESTION.

Enter the number of each question attempted in the box on the front cover of its corresponding answer book.

Hand in 4 answer books even if they have not all been used.

You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.

1. A set of data points $\{x_i, y_i\}_{i=1}^N$ are recorded in an experiment, where x_i are the independent variables and y_i are the dependent variables. A colleague suggests that the relationships between the data should be well described by a second order polynomial of the form,

$$y(x) = a_0 + a_1x + a_2x^2.$$

You are told that the measured values x_i are known perfectly, but the y_i values can each be described as an independent Gaussian random variable with known standard deviation $\delta > 0$,

$$y_i \sim \frac{1}{\delta \sqrt{2\pi}} \exp\left(-\frac{[y_i - y(x_i)]^2}{2\delta^2}\right).$$

for $i = 1, 2, \dots, N$.

- (i) Write down the likelihood function $L(a_0, a_1, a_2)$ for the data points $\{x_i, y_i\}_{i=1}^N$.

ANSWER: SEEN: The likelihood function is given by the product of the probability density functions for each data point,

$$L(a_0, a_1, a_2) = \prod_{i=1}^N \frac{1}{\delta \sqrt{2\pi}} \exp\left(-\frac{(y_i - y(x_i))^2}{2\delta^2}\right).$$

[1] for writing the product over all data points.

[1 mark]

- (ii) Derive a matrix equation that could be solved to find the maximum likelihood estimators $\hat{a}_0, \hat{a}_1, \hat{a}_2$ of the coefficients a_0, a_1, a_2 in terms of the data points $\{x_i, y_i\}_{i=1}^N$. You do not need to solve this equation.

HINT: Start by writing down the negative log-likelihood function: $-\ln L(a_0, a_1, a_2)$.

ANSWER: UNSEEN but extension of the straight line case: The negative log-likelihood function is given by,

$$q(a_0, a_1, a_2) = -\ln L(a_0, a_1, a_2) = \sum_{i=1}^N \left(\frac{(y_i - y(x_i))^2}{2\delta^2} + \ln(\delta \sqrt{2\pi}) \right).$$

[2]. We should get 3 equations for the three estimators by finding $\partial q/\partial a_0$, $\partial q/\partial a_1$ and $\partial q/\partial a_2$ and setting them to zero.

$$\frac{\partial q}{\partial a_0} = 0, \quad \frac{\partial q}{\partial a_1} = 0, \quad \frac{\partial q}{\partial a_2} = 0.$$

[1]. We should notice that each of these will yield a similar expression,

$$\frac{\partial q}{\partial a_j} = \frac{1}{2} \sum_{i=1}^N \frac{1}{\delta^2} \frac{\partial y(x_i)}{\partial a_j} \cdot 2(y(x_i) - y_i)$$

where $j = 0, 1, 2$. We find that,

$$\partial y(x_i)/\partial a_0 = 1, \quad \partial y(x_i)/\partial a_1 = x_i, \quad \partial y(x_i)/\partial a_2 = x_i^2,$$

[This question continues on the next page ...]

[3] and so we can write the matrix equation as,

$$\begin{pmatrix} N & \sum_{i=1}^N x_i & \sum_{i=1}^N x_i^2 \\ \sum_{i=1}^N x_i & \sum_{i=1}^N x_i^2 & \sum_{i=1}^N x_i^3 \\ \sum_{i=1}^N x_i^2 & \sum_{i=1}^N x_i^3 & \sum_{i=1}^N x_i^4 \end{pmatrix} \begin{pmatrix} \hat{a}_0 \\ \hat{a}_1 \\ \hat{a}_2 \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^N y_i \\ \sum_{i=1}^N x_i y_i \\ \sum_{i=1}^N x_i^2 y_i \end{pmatrix}. \quad (1)$$

[3] which is the matrix equation we were asked to derive. Any consistent set of equations is acceptable for the full marks but the entries of the matrix must be written in terms of x_i and y_i otherwise a mark is dropped. [9 marks]

- (iii) Take a look at figure 1 which shows a set of 10 data points $\{x_i, y_i\}_{i=1}^N$ and the best fit second order polynomial $y(x) = \hat{a}_0 + \hat{a}_1 x + \hat{a}_2 x^2$ to the data (solid line) obtained by finding the maximum likelihood estimators. The error bars on each point show the known standard deviation (δ). A second colleague suggests that the relationship

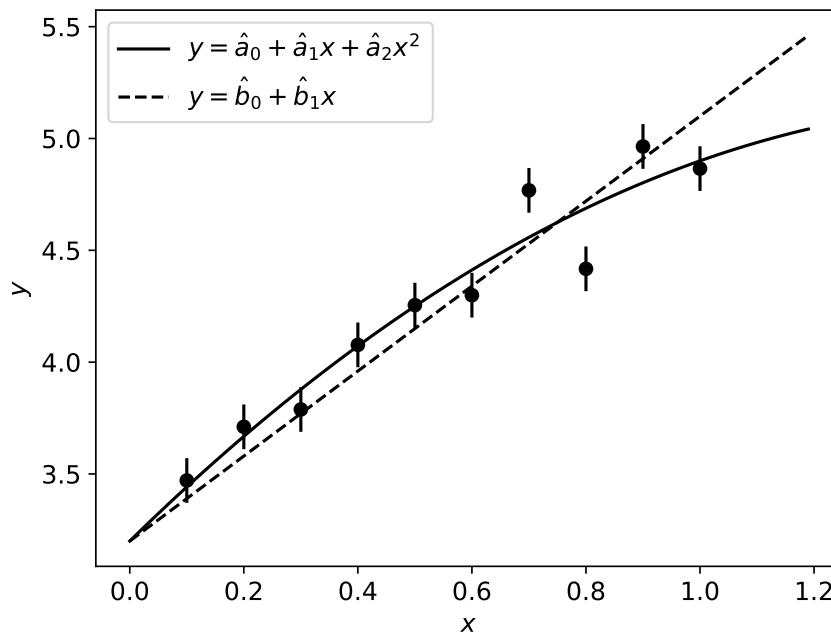


Figure 1

between the y and x values could be better described by a simple linear relationship of the form,

$$y(x) = b_0 + b_1 x.$$

They also calculated the maximum likelihood estimators $\hat{b}_0$, $\hat{b}_1$ for this model and the resulting linear relationship is shown in figure 1 as a dashed line.

- (a) Find the maximum likelihood estimators for the parameters b_0 and b_1 .

HINT: the linear relationship is equivalent to the polynomial relationship when $a_2 = 0$.

ANSWER: SEEN: The maximum likelihood estimators can be obtained by setting $a_0 = b_0$, $a_1 = b_1$ and $a_2 = 0$ in the matrix equation previously obtained. We then have two equations,

$$b_1 \sum_{i=1}^N x_i^2 + b_0 \sum_{i=1}^N x_i - \sum_{i=1}^N y_i x_i = 0$$

and

$$\sum_{i=1}^N y_i - b_1 \sum_{i=1}^N x_i - N b_0 = 0$$

[2]. Notice that $\sum_{i=1}^N y_i = N\bar{y}$ and $\sum_{i=1}^N x_i = N\bar{x}$. The second equation now reads,

$$N\bar{y} - b_1 N\bar{x} - N b_0 = 0 \rightarrow b_0 = \bar{y} - b_1 \bar{x}.$$

Now we substitute the expression for c back into the first equation and we get,

$$b_1 \sum_{i=1}^N x_i^2 + N\bar{x}(\bar{y} - b_1 \bar{x}) - \sum_{i=1}^N x_i y_i = 0$$

which we can rewrite as

$$b_1 \left(\sum_{i=1}^N x_i^2 - N\bar{x}^2 \right) + \left(N\bar{x}\bar{y} - \sum_{i=1}^N x_i y_i \right) = 0$$

and then solve for b_1 to get the maximum likelihood estimator as,

$$\hat{b}_1 = \frac{N\bar{x}\bar{y} - \sum_{i=1}^N x_i y_i}{\sum_{i=1}^N x_i^2 - N\bar{x}^2}.$$

[2]. Then,

$$\hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x} = \bar{y} - \bar{x} \left(\frac{N\bar{x}\bar{y} - \sum_{i=1}^N x_i y_i}{\sum_{i=1}^N x_i^2 - N\bar{x}^2} \right).$$

[2]. These can be expressed in terms of sample means of x and y as above or left with the summation terms. It is also possible to simplify the expression by noticing that,

$$\sum_{i=1}^N x_i^2 - N\bar{x}^2 = \sum_{i=1}^N (x_i - \bar{x})(x_i + \bar{x}), \quad \sum_{i=1}^N x_i y_i - N\bar{x}\bar{y} = - \sum_{i=1}^N (y_i - \bar{y})(x_i + \bar{x}),$$

but this is not needed to get the full marks.

[6 marks]

- (b) Devise a hypothesis test to determine if the best linear model, defined by the maximum likelihood estimates $\hat{b}_0$, $\hat{b}_1$ shown by the dashed line in figure 1, should be rejected in favour of the best second order polynomial model defined

[This question continues on the next

by the maximum likelihood estimates $\hat{a}_0$, $\hat{a}_1$, $\hat{a}_2$ and shown by the solid line in figure 1. You must clearly state the null and alternative hypotheses, the test statistic that you would use, the critical region and size of the test and a full procedure for determining the outcome of the test.

ANSWER: SEEN in different forms: There can be different ways to setup the test but from the lectures, the most powerful would likely be the likelihood ratio test, with the following procedure.

- First, the test statistic is given (using the likelihood ratio) by,

$$t = -2 \ln \frac{L(\hat{b}_0, \hat{b}_1)}{L(\hat{a}_0, \hat{a}_1, \hat{a}_2)}.$$

[1].

- The null hypothesis H_0 should be the linear model and the alternate H_1 is the second order polynomial. [1].
- The size of the test could be anything but let's choose $\alpha = 0.05$ for this example. For this test statistic, we should choose large values of t as the critical region since the second order polynomial should always yield a larger likelihood value - this is because the null hypothesis can be realised by setting $a_2 = 0$ - and so $L(\hat{a}_0, \hat{a}_1, \hat{a}_2) > L(\hat{b}_0, \hat{b}_1)$. [2]
- In order to determine whether to reject H_0 we must calculate the value of t for the data given - t_{obs} , and then calculate the value of $P(t > t_{obs})$. This can be calculating using simulation since we know the distribution of y_i under H_0 at the MLE values. [2].
- If $P(t > t_{obs}) < 0.05$ then we reject H_0 in favour of H_1 , otherwise we fail to reject H_0 . [1].

Other tests are ok, for example a F -test or other test based on the contribution of a_2 or the use of a test statistic that doesn't involve the MLEs of each parameter set is fine. Marks are awarded for reasonable logic in choosing the test and providing a description of all the points above.

[7 marks]

- (c) Define the power $(1 - \beta)$ of a hypothesis test and explain how you would be able to calculate it in your answer to the previous question.

ANSWER: The power is defined as the probability to reject the null hypothesis when the alternative is true. Any of the following definitions would be fine for full marks.

- $1 - \beta = P(t \in w | H_1)$ - the probability of t falling in the critical region w when the alternative is true.
- $1 - \beta = \int_{t_{obs}}^{\infty} f(t | H_1) dt$ - the p -value of the test when the alternative is true.
- $1 - \beta = 1 - P(t \in W - w | H_1)$ - the probability of not making a type-II error.

[This question continues on the next page ...]

[1]. Saying that the power is the probability to reject the null when it is false is not correct as many alternatives could be true instead in this case. The power can be calculated using simulation but this time generated assuming the alternative hypothesis is true **[1]**. [2 marks]

[Total 25 marks]

2. Imagine a telescope measuring the positions of stars in a small region of the sky. The telescope has an unknown position resolution σ that needs to be determined before being put into operation. The telescope is aligned so that the origin $(0,0)$ is centered on the apparent position of the star (shown in figure 1), while the true position is unknown. The X and Y co-ordinates of the difference between the true and apparent positions will be random normal distributed variables with mean 0 and standard deviation σ .

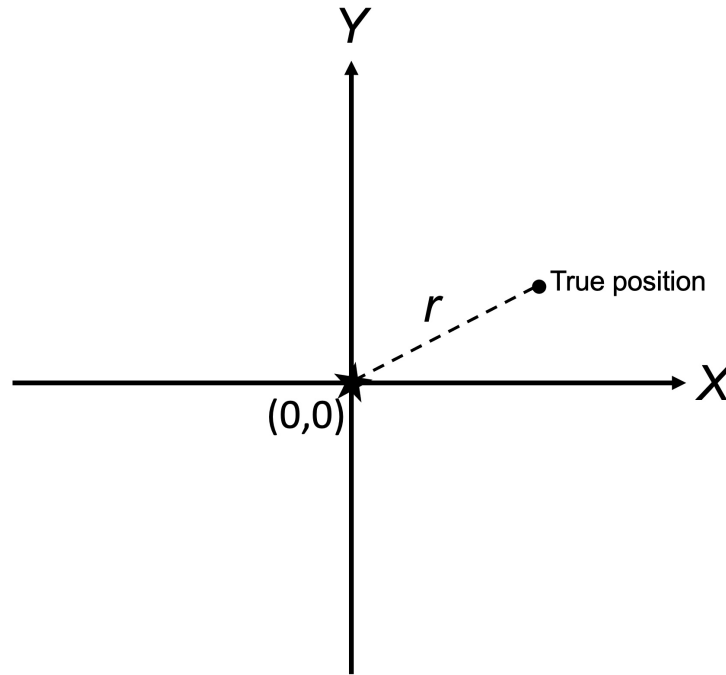


Figure 1

The random variable $r = \sqrt{X^2 + Y^2}$ will be distributed as a Rayleigh distribution, with a probability density given by,

$$f(r; \sigma) = A(\sigma) r e^{-\frac{r^2}{2\sigma^2}},$$

where A is a normalisation term depending on σ . The variable will always be a positive real value $r \in [0, +\infty)$.

- (i) Show that the normalisation term $A(\sigma)$ is given by,

$$A(\sigma) = \frac{1}{\sigma^2}.$$

ANSWER: UNSEEN: The probability distribution should be normalised such that,

$$1 = \int_0^{+\infty} f(r; \sigma) dr.$$

[This question continues on the next page ...]

[1]. Plugging in the Rayleigh distribution, we have,

$$1 = \int_0^{+\infty} A(\sigma) r e^{-\frac{r^2}{2\sigma^2}} dr = A \int_0^{+\infty} \frac{d}{dr} \left(-\sigma^2 e^{-\frac{r^2}{2\sigma^2}} \right) dr = A\sigma^2.$$

Re-arranging, we find that $A(\sigma) = \frac{1}{\sigma^2}$ **[1].** Full marks are also awarded if the expression for A is used to show the integral yields 1. [2 marks]

- (ii) Calculate the second algebraic moment $\mu_2 = E[r^2]$ of the Rayleigh distribution. You may use the following identity,

$$\int_0^{+\infty} x^3 e^{-bx^2} dx = \frac{1}{2b^2}.$$

ANSWER: UNSEEN: The definition of the second algebraic moment is,

$$\mu_2 := E[r^2] = \int_0^{+\infty} r^2 f(r; \sigma) = \frac{1}{\sigma^2} \int_0^{+\infty} r^3 e^{-\frac{r^2}{2\sigma^2}} dr.$$

[1]. Using the provided expression with $b = \frac{1}{2\sigma^2}$ **[1]**, we get,

$$\mu_2 = \frac{1}{\sigma^2} 2\sigma^4 = 2\sigma^2.$$

[1] [3 marks]

- (iii) A set of calibration data $\{r_i\}_{i=1}^N$ using known positions of N stars is used to estimate the resolution of the telescope. These measurements should be assumed to be independent and identically distributed.

- (a) Using the method of moments, find the expression for an estimator of σ^2 using the calibration data from the second algebraic moment.

ANSWER: UNSEEN: In the method of moments, we match the sample moments to moments of the distribution. In this case, we can match the second algebraic moment with the second algebraic sample moment i.e.,

$$\mu_2 = m_2$$

[1]. From the previous part, we find,

$$2\hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N r_i^2.$$

or re-arranging,

$$\hat{\sigma}^2 = \frac{1}{2N} \sum_{i=1}^N r_i^2.$$

[1]. No marks are awarded if the central moments are confused for the algebraic moments. [2 marks]

[This question continues on the next page ...]

- (b) Is the estimator you determined a biased or un-biased estimator?

ANSWER: UNSEEN: The definition of an unbiased estimator $\hat{x}$ for a value x is one for which $E[\hat{x}] - x = 0$ [1]. In our case this means finding,

$$E[\hat{\sigma}^2] - \sigma^2 = E\left[\frac{1}{2N} \sum_{i=1}^N r_i^2\right] - \sigma^2 = \frac{1}{2N} \sum_{i=1}^N E[r_i^2] - \sigma^2 = \frac{N}{2N} E[r^2] - \sigma^2.$$

[1]. We calculated $E[r^2]$ already so we have,

$$E[\hat{\sigma}^2] - \sigma^2 = \frac{N}{2N} 2\sigma^2 - \sigma^2 = 0.$$

[2]. so the estimator is unbiased [1]. [5 marks]

- (c) Write down the likelihood function $L(\sigma)$ for the observations $\{r_i\}_{i=1}^N$.

ANSWER: SEEN in various contexts: We write the likelihood function as,

$$L(\sigma) = \prod_{i=1}^N \frac{r_i}{\sigma^2} e^{-\frac{r_i^2}{2\sigma^2}}$$

[1]. [1 mark]

- (d) Show that the maximum likelihood estimator $\hat{\sigma}$ for σ is given by,

$$\hat{\sigma} = \sqrt{\frac{\sum_{i=1}^N r_i^2}{2N}}.$$

HINT: Start by finding $-\ln L(\sigma)$.

ANSWER: UNSEEN: Taking the negative log of the previous answer yields,

$$-\ln L(\sigma) = -\sum_{i=1}^N \left(-2\ln \sigma + \ln r_i - \frac{r_i^2}{2\sigma^2} \right) = \sum_{i=1}^N \left(2\ln \sigma - \ln r_i + \frac{r_i^2}{2\sigma^2} \right).$$

[1] The minimum is obtained by setting $\frac{d}{d\sigma}(-\ln L(\sigma)) = 0$ [1], hence,

$$0 = \frac{d}{d\sigma}(-\ln L(\sigma)) = \frac{2N}{\hat{\sigma}} - \sum_{i=1}^N \frac{r_i^2}{\hat{\sigma}^3}.$$

[1]. Multiplying through by $\hat{\sigma}$, we get,

$$2N = \sum_{i=1}^N \frac{r_i^2}{\hat{\sigma}^2} \implies \hat{\sigma} = \sqrt{\frac{\sum_{i=1}^N r_i^2}{2N}}$$

[2]. [5 marks]

[This question continues on the next page ...]

- (e) Show that the maximum likelihood estimator $\hat{\sigma}$ is a *biased* estimator of σ . You may use the fact that for $g(x) = \sqrt{x}$, the expectation satisfies $E[g(x)] < g(E[x])$.

ANSWER: UNSEEN: Remembering that the bias of an estimator is the difference between its expectation value and the true value, we consider the expectation of the estimator $\hat{\sigma}$,

$$E[\hat{\sigma}] = E\left[\sqrt{\frac{\sum_{i=1}^N r_i^2}{2N}}\right] = \frac{1}{\sqrt{2N}} E\left[\sqrt{\sum_{i=1}^N r_i^2}\right].$$

[1]. Using the inequality provided, we can move the expectation under the square-root so that,

$$\frac{1}{\sqrt{2N}} E\left[\sum_{i=1}^N r_i^2\right] < \frac{1}{\sqrt{2N}} \sqrt{E\left[\sum_{i=1}^N r_i^2\right]} = \frac{1}{\sqrt{2N}} \sqrt{\sum_{i=1}^N E[r_i^2]} = \frac{1}{\sqrt{2N}} \sqrt{2N}\sigma = \sigma.$$

[2] where we used the result from an earlier question that $E[r^2] = 2\sigma^2$. From this we see that $E[\hat{\sigma}] < \sigma$ so the estimator is biased. **[1]** [4 marks]

- (iv) Aside from being an unbiased estimator, name one other property of a *good* estimator and describe what that property means. You can use equations and/or diagrams in your answer but the explanation must be clearly stated.

ANSWER: SEEN: Any of the following examples from the lectures could be described;

- **Consistency:** An estimator $\hat{\theta}$ is a *consistent* estimator if in the limit of large sample sizes, the estimator converges in probability to the true value θ . This can be written as for all $\epsilon > 0$, $\lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| > \epsilon) = 0$ or some other expression for the limit.
- **Efficiency (or minimum variance):** An estimator $\hat{\theta}$ is an *efficient* estimator if its variance reaches the theoretical minimum set by the Cramer-Rao bound $V(\hat{\theta}) = \frac{1}{I(\theta)}$ where $I(\theta) = E\left[\left(\frac{d(\ln f(X;\theta))}{d\theta}\right)^2\right]$ is the Fisher information. Answers might also be given defining the Fisher information using the second derivative instead. Full marks given if no expression for the Fisher information is given but some mention of the theoretical minimum bound is required.
- **Minimum mean squared error:** Not covered in lectures, but one could say that a good estimator is one that minimises the mean squared error defined by $MS E(\hat{\theta}) = \text{Bias}(\hat{\theta})^2 + V(\hat{\theta})$ which is a combination of the bias and variance. The answer should define each of these terms.
- **Vanishing bias and/or variance:** Again, not seen in lectures but a good estimator can be one for which the limit of the bias and or variance (or the $MS E$) are zero in the limit of large data sample sizes. Again, the terms bias, variance etc must be defined.

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- **Invariance:** A good estimator might be one that is invariant with respect to transformations of the random variable, eg $X \rightarrow X^2 \implies \hat{\theta}(X) = \hat{\theta}(X^2)$ and/or transformations of the parameterisation, eg for some function $\alpha = g(\theta)$, the estimator $\hat{\alpha} = g(\hat{\theta})$.

Other properties may be known but generic properties like “easy to calculate” receive no marks. Talking about error rates or power also receives no marks as that is a confusion with hypothesis testing. One mark is awarded for the name of the property [1] and two marks for a clear explanation of what it means including any equations or diagrams used [2].

[3 marks]

[Total 25 marks]

3. (i) (a) A simple drug trial involves giving N patients the drug and recording how many respond positively to treatment. Assuming that the outcomes of each patient are independent, and the probability of a successful outcome is λ , argue that the sampling distribution for n , the number of patients who respond positively to treatment, is the binomial distribution

$$p(n|N, \lambda) = \frac{N!}{n!(N-n)!} \lambda^n (1-\lambda)^{N-n}.$$

ANSWER: SEEN: The probability of a single successful outcome is λ , and of a failure $1 - \lambda$, so the probability of N independent trials giving n successes in a particular order is $\lambda^n (1 - \lambda)^{N-n}$. [2] Since there are $N!/(n!(N-n)!)$ possible combinations of orderings, we get the result. [2]

[4 marks]

- (b) Write down an unnormalised posterior distribution for λ , given n successful outcomes from a fixed number of patients N . Justify your choice of prior.

ANSWER: VARIANT OF SEEN: $p(\lambda|n) = p(n|\lambda)p(\lambda)/p(n|N) \propto p(n|\lambda) = \frac{N!}{n!(N-n)!} \lambda^n (1-\lambda)^{N-n}$ if we have a uniform prior on λ . Since λ is a location parameter, a uniform prior is suitable. [3]

[3 marks]

- (ii) (a) Find the maximum posterior value for λ for given n and N .

ANSWER: UNSEEN:

$$0 = \frac{\partial p(n|\lambda)}{\partial \lambda} \propto n\lambda^{n-1}(1-\lambda)^{N-n} + (N-n)\lambda^n(1-\lambda)^{N-n-1} \Rightarrow \lambda_{MAP} = \frac{n}{N}.$$

[3]

[3 marks]

- (b) Show that the posterior mean of λ for general n and N is

$$\bar{\lambda} = \frac{n+1}{N+2}.$$

You may assume the following without proof:

$$\int_0^1 x^a (1-x)^b dx = \frac{\Gamma(1+a)\Gamma(1+b)}{\Gamma(2+a+b)},$$

and $\Gamma(z+1) = z\Gamma(z)$.

ANSWER: UNSEEN: The posterior mean is, for a uniform prior,

$$\begin{aligned} \bar{\lambda} &= \frac{\int p(n|\lambda)\lambda d\lambda}{\int p(n|\lambda)d\lambda} \quad [2] \\ &= \frac{\int_0^1 \lambda^{n+1}(1-\lambda)^{N-n} d\lambda}{\int_0^1 \lambda^n(1-\lambda)^{N-n} d\lambda} \quad [2] \\ &= \frac{\Gamma(n+2)\Gamma(N-n+1)}{\Gamma(N+3)} \times \frac{\Gamma(N+2)}{\Gamma(n+1)\Gamma(N-n+1)} \\ &= \frac{n+1}{N+2}. \quad [3] \end{aligned} \quad (1)$$

[This question continues on the next page ...]

[7 marks]

- (iii) (a) What is meant by the term *conjugate prior*?

[2 marks]

ANSWER: A conjugate prior is a prior of a form such that, when multiplied by a likelihood of given form, the resulting posterior has the same form as the prior, but with different parameters. [2]

- (b) Show that the Beta distribution, $\pi(\lambda) \propto \lambda^x(1 - \lambda)^y$ is a conjugate prior for the binomial distribution, and find an expression for the posterior parameters in terms of x, y, n and N .

ANSWER: The posterior in this case is

$$p(\lambda|n, N) \propto p(n|\lambda, N)\pi(\lambda) [2] \propto \frac{N!}{n!(N-n)!} \lambda^n (1-\lambda)^{N-n} \lambda^x (1-\lambda)^y \propto \lambda^{n+x} (1-\lambda)^{N-n+y} [2]$$

which is another Beta distribution with $x \rightarrow x + n, y \rightarrow y + N - n$. [2]

[6 marks]

[Total 25 marks]

4. (i) Write notes on how simulation-based inference (SBI) can be used to estimate Bayesian posteriors, using a simple example with one parameter and one data point for illustration. Using sketches, also show how the sampling distribution may be estimated. Discuss in particular Approximate Bayesian Computation (ABC), and the challenges of SBI.

ANSWER: SEEN: Marks for: SBI conceptually runs many simulations of the data, and keeps only those that match the measured data. The distribution of the parameters of those simulations then estimates the posterior. [2]. SBI deals with situations where the sampling distribution is unknown, and indeed can learn it. [2] Sketch should show a collection of (parameter, data) pairs, with horizontal line representing the posterior and a vertical line the sampling distribution. [2] The challenge is dimensionality (so extreme data compression may be required) and the need for an accurate simulator. [2] Neural network techniques can be used to approximate the distribution of (parameters, data) points (e.g. mixture models, normalising flows etc.) [2]. [10 marks]

- (ii) An experiment consists of 10 independent trials, each of which is either successful or unsuccessful. The number of successes is 6, and the probability of success is a parameter θ that is to be inferred. A simulator draws 100 θ values from a uniform prior between 0 and 1. The parameters and number of successes in a short simulation run is given in the table below. Estimate the posterior, in the form of a histogram in five bins with boundaries at 0, 0.2, 0.4, 0.6, 0.8 and 1.0, using ABC.

[5 marks]

[This question continues on the next page ...]

θ	n	θ	n	θ	n	θ	n
0.25	3	0.22	3	0.18	2	0.54	6
0.97	9	0.05	0	0.34	5	0.37	4
0.80	7	0.81	6	0.46	3	0.75	7
0.61	6	0.39	3	0.66	5	0.07	1
0.97	9	0.87	9	0.78	8	0.82	9
0.81	9	0.49	4	0.77	9	0.57	8
0.25	2	0.71	5	0.76	5	0.18	0
0.61	7	0.99	10	0.33	3	0.77	7
0.85	9	0.69	8	0.69	5	0.83	8
0.91	10	0.40	4	0.26	0	0.14	1
0.74	3	0.59	6	0.37	6	0.45	2
0.31	2	0.40	4	0.90	7	0.28	4
0.93	10	0.42	5	0.39	3	0.01	1
0.75	8	0.51	6	0.84	7	0.34	5
0.64	6	0.73	7	0.75	7	0.95	10
0.45	4	0.65	8	0.73	9	0.71	7
0.94	9	0.80	7	0.22	3	0.87	10
0.30	4	0.05	1	0.07	0	0.11	0
0.55	6	0.73	8	0.71	6	0.89	7
0.94	9	0.38	3	0.20	1	0.94	10
0.32	6	0.71	8	0.53	6	0.07	0
0.44	2	0.62	6	0.05	1	0.65	7
0.36	4	0.09	1	0.41	4	0.78	7
0.47	4	0.16	3	0.37	6	0.29	2
0.46	6	0.56	6	0.65	5	0.63	4

ANSWER: UNSEEN: Selecting only those data that match the measured value (6) gives (0.61,0.64,0.55,0.32,0.46,0.81,0.59,0.51,0.62,0.37,0.71,0.53,0.54). [3]. Construct histogram as instructed [2].

(iii) An important part of SBI is the need for extreme data compression. An experiment records measurements $x_i; i = 1, \dots, N$, drawn independently from a distribution with mean μ , and known gaussian errors σ_i .

(a) Write down the likelihood, and show that it can be factorised into three terms, which are (a) a function of x_i only; (b) a function of μ only; (c) a term that includes both x_i and μ . (σ_i^2 will appear in all of them). [4 marks]

ANSWER: UNSEEN:

$$\begin{aligned}
 L &= \frac{1}{\prod_i \sqrt{2\pi}\sigma_i} \exp\left[-\sum_i \frac{(x_i - \mu)^2}{2\sigma_i^2}\right] [1] \\
 &\propto \exp\left[-\sum_i \frac{x_i^2}{2\sigma_i^2} + \mu \sum_i \frac{x_i}{\sigma_i^2} - \frac{\mu^2}{2} \sum_i \frac{1}{\sigma_i^2}\right] [3]
 \end{aligned} \tag{1}$$

[This question continues on the next page ...]

- (b) Argue that for the posterior of μ , we can ignore the term that depends on x_i only. [3 marks]

ANSWER: UNSEEN: For the posterior, we need to divide by the Bayesian evidence, which is the integral of the likelihood times prior over the parameter(s). Hence a data-only multiplicative term cancels top and bottom. [3].

- (c) By considering the other terms, show that we can compress the x_i into a single number

$$X \equiv \sum_{i=1}^N \frac{x_i}{\sigma_i^2}$$

and still compute the posterior exactly. [3 marks]

ANSWER: UNSEEN: With this cancellation, clearly the posterior depends only on x_i in this combination (or equivalent logic). [3].

[Total 25 marks]